

RUN_ID	
TRAIN_SUBSET_SIZE	
MAX_SOURCE_LENGTH	
MAX_TARGET_LENGTH	
PER_DEVICE_TRAIN_BATCH_SIZE	
PER_DEVICE_EVAL_BATCH_SIZE	
GRADIENT_ACCUMULATION_STEPS	
LEARNING_RATE	
NUM_EPOCHS	
WEIGHT_DECAY	
WARMUP_RATIO	
LABEL_SMOOTHING_FACTOR	
GROUP_BY_LENGTH	
NUM_BEAMS	
MAX_NEW_TOKENS	
LENGTH_PENALTY	
Baseline DEV BLEU score	
RUN DEV BLEU score	
RUN vs Baseline difference	

Interpretation:

RUN_ID	exp01_subset50000_lr2e-5
TRAIN_SUBSET_SIZE	50_000
MAX_SOURCE_LENGTH	128
MAX_TARGET_LENGTH	128
PER_DEVICE_TRAIN_BATCH_SIZE	16
PER_DEVICE_EVAL_BATCH_SIZE	16
GRADIENT_ACCUMULATION_STEPS	1
LEARNING_RATE	2e-5
NUM_EPOCHS	1
WEIGHT_DECAY	0.01
WARMUP_RATIO	0.06
LABEL_SMOOTHING_FACTOR	0.0

GROUP_BY_LENGTH	True
NUM_BEAMS	4
MAX_NEW_TOKENS	128
LENGTH_PENALTY	1.0
Baseline DEV BLEU score	32.96109890363845
RUN DEV BLEU score	32.308711048084845
RUN vs Baseline difference	-0.652388

Interpretation:

Fine-tuning with learning rate at 2e-5 may have been too aggressive on the helsinki model.

RUN_ID	exp02_subset50000_lr2e-5_epoch3
TRAIN_SUBSET_SIZE	50_000
MAX_SOURCE_LENGTH	128
MAX_TARGET_LENGTH	128
PER_DEVICE_TRAIN_BATCH_SIZE	16
PER_DEVICE_EVAL_BATCH_SIZE	16
GRADIENT_ACCUMULATION_STEPS	1
LEARNING_RATE	2e-5
NUM_EPOCHS	3
WEIGHT_DECAY	0.01
WARMUP_RATIO	0.06
LABEL_SMOOTHING_FACTOR	0.0
GROUP_BY_LENGTH	True
NUM_BEAMS	4
MAX_NEW_TOKENS	128
LENGTH_PENALTY	1.0
Baseline DEV BLEU score	32.96109890363845
RUN DEV BLEU score	31.546997647656045
RUN vs Baseline difference	-1.414101

Interpretation:

Increasing the epoch or rerunning the dataset 2 more times made the score significantly worse. We suspect that the model is overfitting to the dataset without proper learning of the sentences. More passes through the dataset made generalization worse.

Next run, we'll go back to NUM_EPOCHS to 1 which yielded a slightly better score, and lower the learning rate to 1e-5

RUN_ID	exp03_subset50000_lr1e-5_epoch1
TRAIN_SUBSET_SIZE	50_000

MAX_SOURCE_LENGTH	128
MAX_TARGET_LENGTH	128
PER_DEVICE_TRAIN_BATCH_SIZE	16
PER_DEVICE_EVAL_BATCH_SIZE	16
GRADIENT_ACCUMULATION_STEPS	1
LEARNING_RATE	1e-5
NUM_EPOCHS	1
WEIGHT_DECAY	0.01
WARMUP_RATIO	0.06
LABEL_SMOOTHING_FACTOR	0.0
GROUP_BY_LENGTH	True
NUM_BEAMS	4
MAX_NEW_TOKENS	128
LENGTH_PENALTY	1.0
Baseline DEV BLEU score	32.96109890363845
RUN DEV BLEU score	32.571798762932985
RUN vs Baseline difference	-0.389300

Interpretation:

By decreasing the learning rate, epoch, and basically the training, we get closer to the baseline score. This shows that the original learning rate of 2e-5 was too aggressive for fine-tuning.

For the next run, we'll decrease the learning rate once more.

RUN_ID	exp04_subset50000_lr1e-6_epoch1
TRAIN_SUBSET_SIZE	50_000
MAX_SOURCE_LENGTH	128
MAX_TARGET_LENGTH	128
PER_DEVICE_TRAIN_BATCH_SIZE	16
PER_DEVICE_EVAL_BATCH_SIZE	16
GRADIENT_ACCUMULATION_STEPS	1
LEARNING_RATE	1e-6
NUM_EPOCHS	1
WEIGHT_DECAY	0.01
WARMUP_RATIO	0.06
LABEL_SMOOTHING_FACTOR	0.0
GROUP_BY_LENGTH	True
NUM_BEAMS	4

MAX_NEW_TOKENS	128
LENGTH_PENALTY	1.0
Baseline DEV BLEU score	32.96109890363845
RUN DEV BLEU score	32.78243481036187
RUN vs Baseline difference	-0.178664

Interpretation:

1e-6 may still be too high or aggressive

For the next run, we'll decrease the learning rate even more

RUN_ID	exp05_subset50000_lr5e-6_epoch1
TRAIN_SUBSET_SIZE	50_000
MAX_SOURCE_LENGTH	128
MAX_TARGET_LENGTH	128
PER_DEVICE_TRAIN_BATCH_SIZE	16
PER_DEVICE_EVAL_BATCH_SIZE	16
GRADIENT_ACCUMULATION_STEPS	1
LEARNING_RATE	5e-6
NUM_EPOCHS	1
WEIGHT_DECAY	0.01
WARMUP_RATIO	0.06
LABEL_SMOOTHING_FACTOR	0.0
GROUP_BY_LENGTH	True
NUM_BEAMS	4
MAX_NEW_TOKENS	128
LENGTH_PENALTY	1.0
Baseline DEV BLEU score	32.96109890363845
RUN DEV BLEU score	32.78789902214486
RUN vs Baseline difference	-0.173200

Interpretation:

Lowering the learning rate continues to show improvement over the previous runs. This suggests that gentle fine-tuning is better than aggressive fine-tuning.

For the next run, we'll decrease the learning rate further.

RUN_ID	exp06_subset50000_lr1e-7_epoch1
TRAIN_SUBSET_SIZE	50_000
MAX_SOURCE_LENGTH	128
MAX_TARGET_LENGTH	128
PER_DEVICE_TRAIN_BATCH_SIZE	16

PER_DEVICE_EVAL_BATCH_SIZE	16
GRADIENT_ACCUMULATION_STEPS	1
LEARNING_RATE	1e-7
NUM_EPOCHS	1
WEIGHT_DECAY	0.01
WARMUP_RATIO	0.06
LABEL_SMOOTHING_FACTOR	0.0
GROUP_BY_LENGTH	True
NUM_BEAMS	4
MAX_NEW_TOKENS	128
LENGTH_PENALTY	1.0
Baseline DEV BLEU score	32.96109890363845
RUN DEV BLEU score	32.916681586057706
RUN vs Baseline difference	-0.044417

Interpretation:

The low learning rate preserves the pretrained model's behavior while making only minimal updates.

For the next run, we'll decrease the learning rate further.

RUN_ID	exp07_subset50000_lr5e-8_epoch1
TRAIN_SUBSET_SIZE	50_000
MAX_SOURCE_LENGTH	128
MAX_TARGET_LENGTH	128
PER_DEVICE_TRAIN_BATCH_SIZE	16
PER_DEVICE_EVAL_BATCH_SIZE	16
GRADIENT_ACCUMULATION_STEPS	1
LEARNING_RATE	5e-8
NUM_EPOCHS	1
WEIGHT_DECAY	0.01
WARMUP_RATIO	0.06
LABEL_SMOOTHING_FACTOR	0.0
GROUP_BY_LENGTH	True
NUM_BEAMS	4
MAX_NEW_TOKENS	128
LENGTH_PENALTY	1.0
Baseline DEV BLEU score	32.96109890363845

RUN DEV BLEU score	32.95748394882813
RUN vs Baseline difference	-0.003615

Interpretation:

This suggests that lowering the learning rate reduces the damage from fine-tuning, but may also make learning too weak to produce a measurable gain

For the next run, we'll decrease the learning rate further.

RUN_ID	exp08_subset50000_lr8e-8_epoch1
TRAIN_SUBSET_SIZE	50_000
MAX_SOURCE_LENGTH	128
MAX_TARGET_LENGTH	128
PER_DEVICE_TRAIN_BATCH_SIZE	16
PER_DEVICE_EVAL_BATCH_SIZE	16
GRADIENT_ACCUMULATION_STEPS	1
LEARNING_RATE	8e-8
NUM_EPOCHS	1
WEIGHT_DECAY	0.01
WARMUP_RATIO	0.06
LABEL_SMOOTHING_FACTOR	0.0
GROUP_BY_LENGTH	True
NUM_BEAMS	4
MAX_NEW_TOKENS	128
LENGTH_PENALTY	1.0
Baseline DEV BLEU score	32.96109890363845
RUN DEV BLEU score	32.99040723366096
RUN vs Baseline difference	0.029308

Interpretation:

This is the first positive, gain score. This goes to show that higher learning rates caused larger drops, while very low learning rates preserved the pretrained model.

For the next run, we'll try increasing the epoch from 1 to 2

RUN_ID	exp09_subset50000_lr8e-8_epoch2
TRAIN_SUBSET_SIZE	50_000
MAX_SOURCE_LENGTH	128
MAX_TARGET_LENGTH	128
PER_DEVICE_TRAIN_BATCH_SIZE	16
PER_DEVICE_EVAL_BATCH_SIZE	16
GRADIENT_ACCUMULATION_STEPS	1

LEARNING_RATE	8e-8
NUM_EPOCHS	2
WEIGHT_DECAY	0.01
WARMUP_RATIO	0.06
LABEL_SMOOTHING_FACTOR	0.0
GROUP_BY_LENGTH	True
NUM_BEAMS	4
MAX_NEW_TOKENS	128
LENGTH_PENALTY	1.0
Baseline DEV BLEU score	32.96109890363845
RUN DEV BLEU score	33.12054867637542
RUN vs Baseline difference	0.159450

Interpretation:

Increasing epochs improved dev SacreBLEU score. With much smaller learning rate, the model could train longer without moving too aggressively away from the pretrained weights.

For the next run, we'll try increasing the epoch from 2 to 3

RUN_ID	exp10_subset50000_lr8e-8_epoch3
TRAIN_SUBSET_SIZE	50_000
MAX_SOURCE_LENGTH	128
MAX_TARGET_LENGTH	128
PER_DEVICE_TRAIN_BATCH_SIZE	16
PER_DEVICE_EVAL_BATCH_SIZE	16
GRADIENT_ACCUMULATION_STEPS	1
LEARNING_RATE	8e-8
NUM_EPOCHS	3
WEIGHT_DECAY	0.01
WARMUP_RATIO	0.06
LABEL_SMOOTHING_FACTOR	0.0
GROUP_BY_LENGTH	True
NUM_BEAMS	4
MAX_NEW_TOKENS	128
LENGTH_PENALTY	1.0
Baseline DEV BLEU score	32.96109890363845
RUN DEV BLEU score	33.14745419426738
RUN vs Baseline difference	0.186355

Interpretation:

The improvement gain was smaller than the jump from 1 to 2 epochs.

For the next run, we'll try increasing the epoch from 3 to 4

RUN_ID	exp11_subset50000_lr8e-8_epoch4
TRAIN_SUBSET_SIZE	50_000
MAX_SOURCE_LENGTH	128
MAX_TARGET_LENGTH	128
PER_DEVICE_TRAIN_BATCH_SIZE	16
PER_DEVICE_EVAL_BATCH_SIZE	16
GRADIENT_ACCUMULATION_STEPS	1
LEARNING_RATE	8e-8
NUM_EPOCHS	4
WEIGHT_DECAY	0.01
WARMUP_RATIO	0.06
LABEL_SMOOTHING_FACTOR	0.0
GROUP_BY_LENGTH	True
NUM_BEAMS	4
MAX_NEW_TOKENS	128
LENGTH_PENALTY	1.0
Baseline DEV BLEU score	32.96109890363845
RUN DEV BLEU score	33.02260431476447
RUN vs Baseline difference	0.061505

Interpretation:

The score dropped compared to the previous run of 3 epochs. Too many passes over the 50,000 subset can hurt generalization.

For the next run, we'll drop epoch from 4 to 3, and remove weight_decay

RUN_ID	exp12_subset50000_lr8e-8_epoch3_wd0
TRAIN_SUBSET_SIZE	50_000
MAX_SOURCE_LENGTH	128
MAX_TARGET_LENGTH	128
PER_DEVICE_TRAIN_BATCH_SIZE	16
PER_DEVICE_EVAL_BATCH_SIZE	16
GRADIENT_ACCUMULATION_STEPS	1
LEARNING_RATE	8e-8
NUM_EPOCHS	3
WEIGHT_DECAY	0.0

WARMUP_RATIO	0.06
LABEL_SMOOTHING_FACTOR	0.0
GROUP_BY_LENGTH	True
NUM_BEAMS	4
MAX_NEW_TOKENS	128
LENGTH_PENALTY	1.0
Baseline DEV BLEU score	32.96109890363845
RUN DEV BLEU score	33.14745419426738
RUN vs Baseline difference	0.186355

Interpretation

This shows that weight_decay did not affect the score because the learning rate was already extremely small.

For the next run, we'll revert weight_decay, and increase label smoothing factor

RUN_ID	exp13_subset50000_lr8e-8_epoch3_lsf0.1
TRAIN_SUBSET_SIZE	50_000
MAX_SOURCE_LENGTH	128
MAX_TARGET_LENGTH	128
PER_DEVICE_TRAIN_BATCH_SIZE	16
PER_DEVICE_EVAL_BATCH_SIZE	16
GRADIENT_ACCUMULATION_STEPS	1
LEARNING_RATE	8e-8
NUM_EPOCHS	3
WEIGHT_DECAY	0.01
WARMUP_RATIO	0.06
LABEL_SMOOTHING_FACTOR	0.1
GROUP_BY_LENGTH	True
NUM_BEAMS	4
MAX_NEW_TOKENS	128
LENGTH_PENALTY	1.0
Baseline DEV BLEU score	32.96109890363845
RUN DEV BLEU score	33.24900519591735
RUN vs Baseline difference	0.287906

Interpretation:

The label smoothing helped regularize fine-tuning and improved generalization to the development set. Because machine translation often have multiple valid outputs, label smoothing prevents the model from becoming too overconfident that only the exact reference wording is correct.

For the next run, we'll decrease the label smoothing factor

RUN_ID	exp14_subset50000_lr8e-8_epoch3_lsf0.05
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TRAIN_SUBSET_SIZE	50_000
MAX_SOURCE_LENGTH	128
MAX_TARGET_LENGTH	128
PER_DEVICE_TRAIN_BATCH_SIZE	16
PER_DEVICE_EVAL_BATCH_SIZE	16
GRADIENT_ACCUMULATION_STEPS	1
LEARNING_RATE	8e-8
NUM_EPOCHS	3
WEIGHT_DECAY	0.01
WARMUP_RATIO	0.06
LABEL_SMOOTHING_FACTOR	0.05
GROUP_BY_LENGTH	True
NUM_BEAMS	4
MAX_NEW_TOKENS	128
LENGTH_PENALTY	1.0
Baseline DEV BLEU score	32.96109890363845
RUN DEV BLEU score	33.14452514096906
RUN vs Baseline difference	0.183426

Interpretation:

A smaller value of 0.05 did not improve over the no-smoothing configuration, so the benefit appears to come from a stronger regularization effect

For the next run, we'll increase the label smoothing factor from our best

RUN_ID	exp15_subset50000_lr8e-8_epoch3_lsf0.2
TRAIN_SUBSET_SIZE	50_000
MAX_SOURCE_LENGTH	128
MAX_TARGET_LENGTH	128
PER_DEVICE_TRAIN_BATCH_SIZE	16
PER_DEVICE_EVAL_BATCH_SIZE	16
GRADIENT_ACCUMULATION_STEPS	1
LEARNING_RATE	8e-8
NUM_EPOCHS	3
WEIGHT_DECAY	0.01
WARMUP_RATIO	0.06
LABEL_SMOOTHING_FACTOR	0.2
GROUP_BY_LENGTH	True

NUM_BEAMS	4
MAX_NEW_TOKENS	128
LENGTH_PENALTY	1.0
Baseline DEV BLEU score	32.96109890363845
RUN DEV BLEU score	33.2804318949317
RUN vs Baseline difference	0.319333

Interpretation:

Increasing label smoothing to 0.2 improved dev SacreBLEU score from our best. This suggests that stronger regularization is helping the fine-tuned model generalize better.

For the next run, we'll increase the label smoothing factor from our best

RUN_ID	exp16_subset50000_lr8e-8_epoch3_lsf0.3
TRAIN_SUBSET_SIZE	50_000
MAX_SOURCE_LENGTH	128
MAX_TARGET_LENGTH	128
PER_DEVICE_TRAIN_BATCH_SIZE	16
PER_DEVICE_EVAL_BATCH_SIZE	16
GRADIENT_ACCUMULATION_STEPS	1
LEARNING_RATE	8e-8
NUM_EPOCHS	3
WEIGHT_DECAY	0.01
WARMUP_RATIO	0.06
LABEL_SMOOTHING_FACTOR	0.3
GROUP_BY_LENGTH	True
NUM_BEAMS	4
MAX_NEW_TOKENS	128
LENGTH_PENALTY	1.0
Baseline DEV BLEU score	32.96109890363845
RUN DEV BLEU score	33.20137289600071
RUN vs Baseline difference	0.240274

Interpretation:

This scored lower than label smoothing factor at 0.2. This suggests that 0.3 may be too smooth. The model becomes less confident in the correct target tokens, which can weaken translation quality

For the next run, we'll revert back to our best and change batch size

RUN_ID	exp17_subset50000_lr8e-8_epoch3_lsf0.2_bs32
TRAIN_SUBSET_SIZE	50_000
MAX_SOURCE_LENGTH	128

MAX_TARGET_LENGTH	128
PER_DEVICE_TRAIN_BATCH_SIZE	32
PER_DEVICE_EVAL_BATCH_SIZE	32
GRADIENT_ACCUMULATION_STEPS	1
LEARNING_RATE	8e-8
NUM_EPOCHS	3
WEIGHT_DECAY	0.01
WARMUP_RATIO	0.06
LABEL_SMOOTHING_FACTOR	0.2
GROUP_BY_LENGTH	True
NUM_BEAMS	4
MAX_NEW_TOKENS	128
LENGTH_PENALTY	1.0
Baseline DEV BLEU score	32.96109890363845
RUN DEV BLEU score	33.31063952348674
RUN vs Baseline difference	0.349541

Interpretation:

Increasing the batch size from 16 to 32 improved dev SacreBLEU. This suggests that a larger batch gave a more stable gradient estimate and helped fine-tuning generalize slightly better. It processed more examples per step.

For the last run, we'll use our best configuration and increase num_beams to 5

RUN_ID	exp18_subset50000_lr8e-8_epoch3_lsf0.2_bs32_beams5
TRAIN_SUBSET_SIZE	50_000
MAX_SOURCE_LENGTH	128
MAX_TARGET_LENGTH	128
PER_DEVICE_TRAIN_BATCH_SIZE	32
PER_DEVICE_EVAL_BATCH_SIZE	32
GRADIENT_ACCUMULATION_STEPS	1
LEARNING_RATE	8e-8
NUM_EPOCHS	3
WEIGHT_DECAY	0.01
WARMUP_RATIO	0.06
LABEL_SMOOTHING_FACTOR	0.2
GROUP_BY_LENGTH	True
NUM_BEAMS	5
MAX_NEW_TOKENS	128

LENGTH_PENALTY	1.0
Baseline DEV BLEU score	32.96109890363845
RUN DEV BLEU score	33.38585783968027
RUN vs Baseline difference	0.424759

Interpretation:

This decoding/generation hyperparameter experiment saw an increase without additional training. This means the trained model already had better translation candidates available, and the slightly larger beam search helped select better output sequences during generation.